21.8.1 Bio-predators

These are naturally occurring insects that use pests for feeding or multiplication. These are called bio-predators. By introducing large numbers of predators in a greenhouse we can destroy the pest. Predators like Chrysopa spp. and Menochilus spp. are highly useful in controlling pests like aphids, white flies, cotton bollworms, leaf insects etc.

DO YOU KNOW?

The black kneel capsid is an insect found on fruit trees. It eats more than 1000 fruit tree red spider mites per year.



Figure 21.25 Predatory Lady bird beetle feeding on Aphid

21.8.2 Bio-pesticide

Bio-pesticides are living organism or their derived parts which are used as bio-control agents to protect crops against insect pests. Bio-pesticides are of different types based on their origin.

a. Fungal bio-pesticides

Trichodermaviride is a fungus used as a biological pesticide. It is useful to control various disease caused by fungi such as wilt, rusting of leaves and root disease.

b. Bacterial bio-pesticide

A culture of bacillus thuringiensis bacteria is effectively used to control the pest Lepidoptera that attack cotton and maize plants. Panchagavya and leaves decoction of some plants are also used as bio- pesticides.

21.8.3 Bio-repellant

Compound Azadiractin obtained from seeds of neem serves as a good insect-repellant. One of the earliest pesticides used by man was margosa leaves. The dried leaves repel the pests from stored grains.

21.8.4 Bio-fertilizer

Bio-fertilizers are organisms which can bring about soil nutrient enrichment. Nitrogen fixing microorganisms have the capability of converting free nitrogen into nitrogenous compounds and make the soil fertile. The main source of bio-fertilizers is cyano bacteria and certain fungi. Free living cyano bacterium involves in nitrogen fixation along with photosynthesis. Eg. Anabeana, Nostoc. Symbiotic bacteria also fix atmospheric nitrogen. E.g. Rhizobium. Although the chemical fertilizers increase food production, they degrade the natural habitat.



Figure 21.26 Bio-fertilizer

Activity 7

Take a leguminous plant like pea and find out if there are any nodes. Rhizobium bacteria live in such nodes.



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